7 NON-EQUILIBRIUM EXCESS CARRIER

When an external excitation is applied to a semiconductor, excess electrons in conduction band and excess holes in valence band will exist in addition to the thermal-equilibrium concentrations.

Any deviation from thermal equilibrium will tend to change the electron and hole concentrations in a semiconductor.

Example of external excitation are optical illumination and external bias.

- In section 3.3 we have discussed that optical illumination with photon energy $E_{ph} \geq E_g$ can give rise to excess electrons and holes.
- External bias can either inject electrons, holes or both into semiconductor
- Excess electrons and holes do not move independently of each other

- They diffuse, drift and recombine with the same effective diffusion coefficient, drift mobility and lifetime.
  - For an extrinsic semiconductor where the thermal equilibrium majority concentration is much larger than the minority concentration and under low level injection which means the excess carrier concentration is much less than the thermal equilibrium majority concentration, the effective diffusion, mobility and lifetime are those of the minority carrier (for detail analysis and justification, refer to: Donald A. Neamen, "Semiconductor Physics and Devices Basic Principle", 3rd edition McGraw-Hill 2003, Chapter 6)
- Behavior of excess carriers is fundamental to operation of semiconductor devices

7.1 Excess Carrier Concentrations (Direct Band-to-Band Transitions)

To analyze the deviation from thermal equilibrium situation, let us start with the equilibrium situation discussed in section 3.3 which we review here.

- Electrons are continually being thermally excited from valence band into conduction band by thermal energy.
- At the same time, electrons moving randomly through crystal in conduction band may come in close proximity to holes and "fall" into empty states in valence band. In this process the electrons and holes are annihilated (i.e. disappear). This process is called recombination. The processes are illustrated in Fig. 7.1

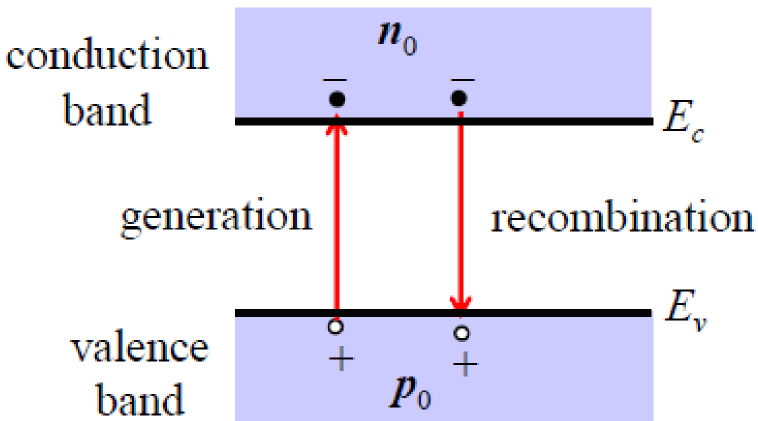


Fig. 7.1

- In thermal equilibrium:
  - Thermal generation rate G_{th} = recombination rate R **(7.1)**

    where G_{th} is the number of electrons and holes generated thermally per unit volume per second (units: #/cm^3-s)

    and R is the number of electrons and holes recombining per unit volume per second (units: #/cm^3-s)

  - There exist thermal equilibrium electron and hole concentrations, n_0 and p_0 respectively.

- The direct generation and recombination processes from band-to-band shown in Fig. 7.1 is called direct band-to-band generation and recombination.

- In the direct band-to-band recombination that we are considering here, electrons and holes recombine directly (i.e. not via traps or recombination centers in the bandgap) and spontaneously. We therefore expect that the recombination rate R is proportional to the electron concentration and hole concentration.

  Consequently, we can write a general expression for the recombination rate R as

$$\boxed{R = \alpha_r \, n \, p} \qquad \textbf{(7.2)}$$

  where α_r is a constant of proportionality for recombination.

- At thermal equilibrium the electron and hole concentrations are n_0 and p_0 respectively. Applying eqn. (7.2) to the thermal equilibrium case [(eqn. 7.1)] we obtain the thermal generation rate G_{th},

$$\boxed{G_{th} = R = \alpha_r\, n\, p = \alpha_r\, n_0\, p_0 = \alpha_r\, n_i^2} \quad \textbf{(7.3)}$$

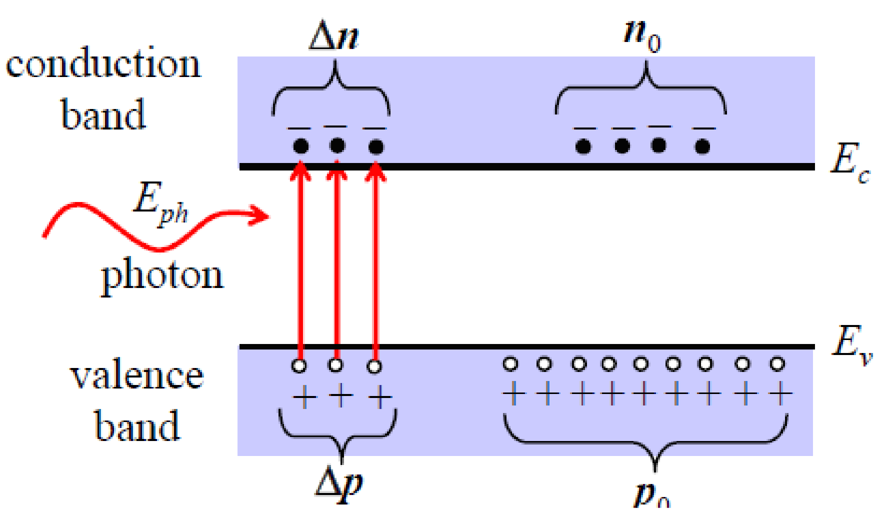
[From eqn. (5.31): $n_0 p_0 = n_i^2$]

Let us now consider a case where a semiconductor sample is subjected to an optical illumination with $E_{ph} \geq E_g$.

- The optical illumination will excite electrons in the valence band to the conduction band creating additional electron-hole pairs. This is shown schematically in Fig. 7.2.

  The additional electrons and holes created are called excess electrons and excess holes.

Fig. 7.2



- With the excess electrons and holes, the electron and hole concentration is

$$\boxed{\begin{aligned} n &= n_0 + \Delta n \\ p &= p_0 + \Delta p \end{aligned}} \qquad \begin{aligned} &\textbf{(7.4.a)} \\ &\textbf{(7.4.b)} \end{aligned}$$

  where Δn and Δp are the excess electron and hole concentrations respectively

- For direct band-to-band transitions, the electrons and holes are generated and recombined in pairs. Therefore, in this case

$$\Delta n = \Delta p \qquad \textbf{(7.5)}$$

- Since the semiconductor is no longer in thermal equilibrium, from eqn. (7.4) we get

$$np \neq n_0 p_0 = n_i^2 \qquad \textbf{(7.6)}$$

Our task now is to evaluate the excess electron and hole concentrations under the influence of the external excitation, in our case here the optical illumination.

For the reason given earlier, we will consider the excess minority carrier concentration, although for the band to band transitions $\Delta n = \Delta p$.

- Consider a p-type semiconductor with $p_0 >> n_0$. The minority carrier is electron.

  In term of generation rate G and the recombination rate R, we can write a general expression for the net rate of change of the minority carrier concentration, that is

$$\frac{dn}{dt} = \text{generation rate } G - \text{recombination rate } R \quad \textbf{(7.7)}$$

The generation rate G is the total generation rate, i.e. the sum of the thermal generation rate and the generation rate by an external excitation.

So, we can write

$$G = G_{th} + G_L \qquad \textbf{(7.8)}$$

where G_{th} is the thermal generation rate, that is the number of electrons and holes generated per unit volume per second by thermal energy (units #/cm^3-s)

and G_L is the number of electrons and holes generated per unit volume per second by the external excitation, in our case here optical excitation, (units #/cm^3-s)

- Using eqn. (7.8), eqn. (7.7) becomes

$$\frac{dn}{dt} = G_{th} + G_L - R \qquad \textbf{(7.9)}$$

Eqn. (7.9) is a general equation which can be applied to various circumstances. Here, we will apply eqn. (7.9) to three different case, namely

I. Thermal equilibrium

II. Steady state condition under an optical illumination

III. State after the illumination is switched off.

I. Thermal equilibrium case

Recall eqn. (7.9): $$\frac{dn}{dt} = G_{th} + G_L - R$$

At thermal equilibrium

- The electron and hole concentrations are constant with time, this implies

$$\frac{dn}{dt} = 0 \qquad \textbf{(7.10)}$$

- There is no external excitation, this implies

$$G_L = 0 \qquad \textbf{(7.11)}$$

Using eqns. (7.10) and (7.11), eqn. (7.9) becomes

$$G_{th} = R$$

At thermal equilibrium the electron and hole concentrations are n_0 and p_0 respectively. Using eqn. (7.2), we then obtain

$$\boxed{G_{th} = R = \alpha_r \, n \, p = \alpha_r \, n_0 \, p_0 = \alpha_r \, n_i^{\,2}}$$

This is exactly what we have obtained in eqn. (7.3).

$$\boxed{G_{th} = \mathrm{R} = \frac{p_0}{\tau_{p0}} = \frac{n_0}{\tau_{n0}}}$$

$$G_{th} = \frac{minority\ carrier}{minority\ carrier\ life\ time}$$

$$= \frac{majority\ carrier}{majority\ carrier\ life\ time}$$

where

$$\tau_{p0} = \frac{1}{\alpha_r n_0} \quad \text{and} \quad \tau_{n0} = \frac{1}{\alpha_r p_0}$$

II. Steady state condition under an optical illumination

Consider p-type semiconductor and start again from eqn. (7.9):

$$\frac{dn}{dt} = G_{th} + G_L - R$$

- At the steady state the electron and hole concentrations are constant with time, therefore

$$\frac{dn}{dt} = 0 \qquad \textbf{(7.12)}$$

- This implies the total generation rate is equal the recombination rate.

  Eqn. (7.9) becomes

$$G_{th} + G_L - R = 0 \qquad \textbf{(7.13)}$$

Rearrange eqn. (7.13), we write

$$G_L = -\left(G_{th} - R\right)$$

- Using eqn. (7.3) for G_{th}, eqn. (7.2) for R and eqn. (7.4) for n and p, we obtain

$$G_L = -\alpha_r \left(n_i^{\,2} - np\right)$$

$$\begin{aligned} G_L &= -\alpha_r \left(n_i^{\,2} - (n_0 + \Delta n)(p_0 + \Delta p)\right) \\ &= -\alpha_r \left(n_i^2 - (n_0 p_0 + n_0 \Delta p + p_0 \Delta n + \Delta n \Delta p)\right) \\ &= \alpha_r \left(n_0 \Delta p + p_0 \Delta n + \Delta n \Delta p\right) \qquad \textbf{(7.14)} \end{aligned}$$

- Since excess electrons and holes are created and recombine in pairs, we have $\Delta n = \Delta p$ (see eqn. 7.5)

  We can then write eqn. (7.14) as follow

$$G_L = \alpha_r \, \Delta n \left(n_0 + p_0 + \Delta n \right) \qquad \textbf{(7.15)}$$

- For a p-type semiconductor, we can easily solve eqn. (7.15) for the following condition

  - $p_0 >> n_0$ **(7.16)**

  - low-level injection which means the excess carrier concentration is much less than the thermal equilibrium majority carrier concentration, which means

$$\Delta n << p_0 \qquad \textbf{(7.17)}$$

- Hence, eqn. (7.15) becomes

$$\boxed{G_L = \alpha_r\, p_0\, \Delta n = \frac{\Delta n}{\tau_{n0}}} \quad \textbf{(7.18)}$$

$$G_L = \frac{excess\ minority\ carrier}{minority\ carrier\ life\ time}$$

where $$\boxed{\tau_{n0} = \frac{1}{\alpha_r\, p_0}} \quad (7.19)$$

τ_{n0} is often referred to as the excess minority carrier lifetime or just minority carrier lifetime.

Similarly, for n-type semiconductors with

- $n_0 >> p_0$ **(7.20)**

- and under low level injection $\Delta p << n_0$ **(7.21)**

we obtain

$$\boxed{G_L = \alpha_r\, n_0\, \Delta p = \frac{\Delta p}{\tau_{p0}}} \qquad (7.22)$$

$$G_L = \frac{\textit{excess minority carrier}}{\textit{minority carrier life time}}$$

where

$$\boxed{\tau_{p0} = \frac{1}{\alpha_r\, n_0}} \qquad \mathbf{(7.23)}$$

Again, τ_{p0} is also referred to as the excess minority carrier lifetime or just minority carrier lifetime.

We can rewrite eqns. (7.18) and (7.22) to obtain the excess carrier concentrations as follow

- p-type semiconductor $$\boxed{\Delta n = \Delta p = G_L \, \tau_{n0}} \quad \textbf{(7.24.a)}$$

- n-type semiconductor $$\boxed{\Delta n = \Delta p = G_L \, \tau_{p0}} \quad \textbf{(7.24.b)}$$

Note that in eqns. (7.24.a) and (7.24.b) we have used $\Delta n = \Delta p$, because for the case we are considering here the electrons and holes are generated and recombined in pairs (direct band to band transitions)

It should also be pointed out here that, in general, the generation and recombination rates may be functions of space coordinates and time

III State after the optical illumination is switched off

At the steady state condition under an optical illumination, we have excess carrier concentrations, Δn and Δp, described in eqns. (7.24)

If we now switch off the illumination, we expect that the excess carrier concentrations would decay and after a long time ($t = \infty$) the electron and hole concentration would return to their thermal equilibrium value n_0 and p_0. The decay of the excess carrier concentration is illustrated in Fig. 7.3.

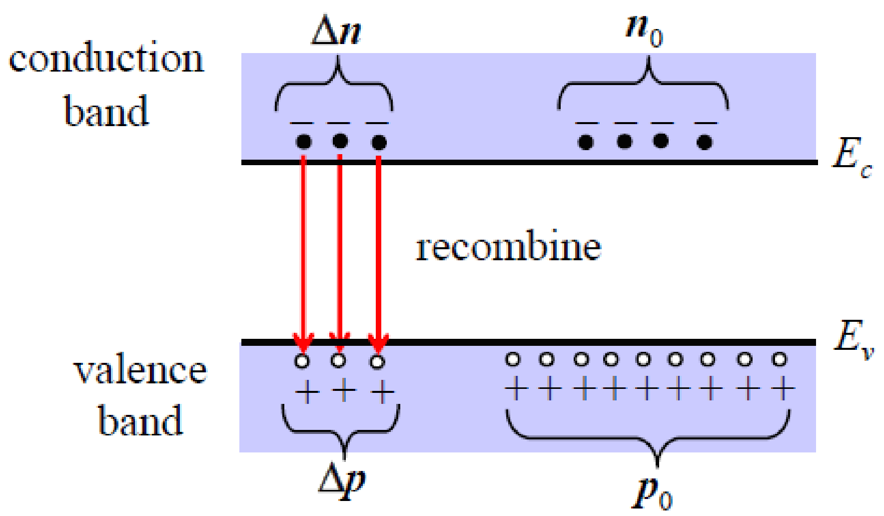


Fig. 7.3

We will now analyse the excess carrier concentrations, Δn and Δp, as a function of time t after the optical illumination is switched off. We will again consider the minority carrier concentration.

- Consider a p-type semiconductor with $p_0 >> n_0$. As before, we start with eqn. (7.9)

$$\frac{dn}{dt} = G_{th} + G_L - R$$

- After the optical illumination is switched off, we define at time $t = 0$, there is no electrons and holes generated by the external excitation, therefore

$$G_L = 0 \qquad \textbf{(7.25)}$$

The electrons and hole are only generated by the thermal energy.

- Hence eqn. (7.9) becomes

$$\frac{dn}{dt} = G_{th} - R \qquad \textbf{(7.26)}$$

dn/dt is eqn. (7.26) is not equal to zero, since the state is not in an equilibrium or a steady state condition. The excess electrons and holes would recombine decaying with time. The electron and hole concentrations are not constant with time.

We can evaluate the excess carrier concentrations as a function of time by solving eqn. (7.26). As we will see, we will carry out the analysis in much similar way as we previously did when we solved the steady state condition.

- Using eqn. (7.3) for G_{th}, eqn. (7.2) for R and eqn. (7.4) for n and p, eqn. (7.26) becomes

$$\frac{dn}{dt} = \alpha_r\left(n_i^2 - np\right)$$

$$\frac{dn}{dt} = \alpha_r\left(n_i^2 - (n_0 + \Delta n)(p_0 + \Delta p)\right)$$

$$= \alpha_r\left(n_i^2 - (n_0 p_0 + n_0\Delta p + p_0\Delta n + \Delta n\Delta p)\right)$$

$$\frac{dn}{dt} = -\alpha_r\left(n_0\Delta p + p_0\Delta n + \Delta n\Delta p\right) \qquad \textbf{(7.27)}$$

Since the electrons and holes are generated and recombined in pairs, i.e. $\Delta n = \Delta p$, eqn. (7.27) can be written as

$$\frac{dn}{dt} = -\alpha_r\, \Delta n\left(n_0 + p_0 + \Delta n\right) \qquad \textbf{(7.28)}$$

- As before, for a p-type semiconductor that we are considering, we solve eqn. (7.28) for the following condition
  - $p_0 >> n_0$
  - low-level injection $\Delta n << p_0$

We then obtain

$$\frac{dn}{dt} = -\alpha_r \, p_0 \, \Delta n = -\frac{\Delta n}{\tau_{n0}} \quad \textbf{(7.29)} \qquad \text{where} \qquad \tau_{n0} = \frac{1}{\alpha_r \, p_0} \quad \textbf{(7.30)}$$

Note that τ_{n0} in eqn. (7.30) is the same as that in eqn. (7.19)

- From eqn. (7.4.a) we can write eqn. (7.29) as follow

$$\frac{dn}{dt}=\frac{d\left(n_0+\Delta n\right)}{dt}=\frac{d\Delta n}{dt} \qquad \textbf{(7.31)}$$

Hence, eqn. (7.29) becomes $$\frac{d\Delta n}{dt}=-\frac{\Delta n}{\tau_{n0}} \qquad \textbf{(7.32)}$$

- The solution of eqn. (7.32) is an exponential decay from the initial excess concentration

$$\boxed{\Delta n(t)=\Delta n(0)\exp\left(-\frac{t}{\tau_{n0}}\right)} \qquad \textbf{(7.33)}$$

where $\Delta n(0)$ is the excess minority carrier electron concentration at $t = 0$ or the initial excess minority carrier electron concentration

- For an optical excitation with generation rate G_L, we can use eqn. (7.24.a) to obtain $\Delta n(0)$

$$\Delta n(0) = G_L \ \tau_{n0} \qquad \textbf{(7.34)}$$

- For $t = \infty$, the excess carrier concentration $\Delta n = 0$. This is expected since long time after the optical excitation is switched off (i.e. $t = \infty$), the carrier concentrations would return to their thermal equilibrium value n_0 and p_0.

Similarly, for n-type semiconductors with

- $n_0 >> p_0$
- and under low level injection $\Delta p << n_0$

We obtain $$\frac{dp}{dt} = \frac{d\Delta p}{dt} = G_{th} - R = \alpha_r\left(n_i^2 - np\right) = -\frac{\Delta p}{\tau_{p0}} \quad \textbf{(7.35)}$$

hence $$\boxed{\Delta p(t) = \Delta p(0)\exp\left(-\frac{t}{\tau_{p0}}\right)} \quad \textbf{(7.36)}$$

where $\Delta p(0)$ is the excess minority carrier hole concentration at $t = 0$ or the initial excess minority carrier hole concentration

and $\tau_{p0} = \dfrac{1}{\alpha_r\, n_0}$ is the same as that in eqn. (7.23)

- From the above discussion, it is clear that the excess electrons and holes recombine at the same rate determined by the minority carrier lifetime
    - For p-type semiconductor by the minority carrier electron lifetime
    - For n-type semiconductor by the minority carrier hole lifetime.

Example 7.1

Consider a piece of p-type semiconductor sample at 300 K doped to concentration $N_a = 10^{15}/cm^3$. The intrinsic carrier concentration is $1.5\times10^{10}/cm^3$. Assume that 10^{14} electron-hole pairs per cm^3 have been created and exist in the sample for $t < 0$. The minority carrier lifetime is 10 ns. With the external excitation removed for $t \geq 0$, calculate the excess carrier concentrations for $t = 10$ ns, 20 ns, 30 ns and ∞.

Plot the carrier concentrations versus time showing the decay of the excess carrier concentration.

Comment on the percentage change of the minority and majority carrier concentration

For complete ionization (300 K) and $N_a >> n_i$, the majority carrier concentration $p_0 = N_a = 10^{15}/\text{cm}^3$.

The initial excess carrier concentrations $\Delta n(0) = \Delta p(0) = 10^{14}/\text{cm}^3 = 0.1\ p_0$.

We may assume low level injection applies. Therefore, from eqn. (7.33) , the excess minority carrier electron concentration is

$$\Delta n(t) = \Delta n(0)\exp\left(-\frac{t}{\tau_{n0}}\right)$$

with initial excess carrier concentration $\Delta n\ (0) = \Delta p\ (0) = 10^{14}/\text{cm}^3$.

For direct band-to-band transitions, at any time t

$$\Delta n\ (t) = \Delta p\ (t)$$

Therefore, for $\boldsymbol{t}$ **= 10 ns**

$$\Delta n(10\text{ ns}) = \Delta p(10\text{ ns}) = 10^{14}\exp\left(-\frac{10\times10^{-9}}{10\times10^{-9}}\right) = 3.7\times10^{13}/\text{cm}^3$$

for $\boldsymbol{t}$ **= 20 ns**

$$\Delta n(20\text{ ns}) = \Delta p(20\text{ ns}) = 10^{14}\exp\left(-\frac{20\times10^{-9}}{10\times10^{-9}}\right) = 1.35\times10^{13}/\text{cm}^3$$

for $\boldsymbol{t}$ **= 30 ns**

$$\Delta n(30\text{ ns}) = \Delta p(30\text{ ns}) = 10^{14}\exp\left(-\frac{30\times10^{-9}}{10\times10^{-9}}\right) = 4.98\times10^{12}/\text{cm}^3$$

for $\boldsymbol{t} = \infty$

$$\Delta n(\infty) = \Delta p(\infty) = 10^{14}\exp\left(-\frac{\infty}{10\times10^{-9}}\right) = 0$$

The carrier concentrations return to their thermal equilibrium values

Fig. 7.4 shows the decay of excess electron and hole concentrations by direct band-to-band recombination

- initial excess carrier concentrations $\Delta n(0) = \Delta p(0) = 10^{14}/\text{cm}^3 = 0.1\ p_0$
- $p_0 = 10^{15}/\text{cm}^3$ and n_0 is negligible
- minority carrier lifetime $\tau_{n0} = 10$ ns

The exponential decay of $\Delta n(t)$ is linear on this semi-logarithmic graph.

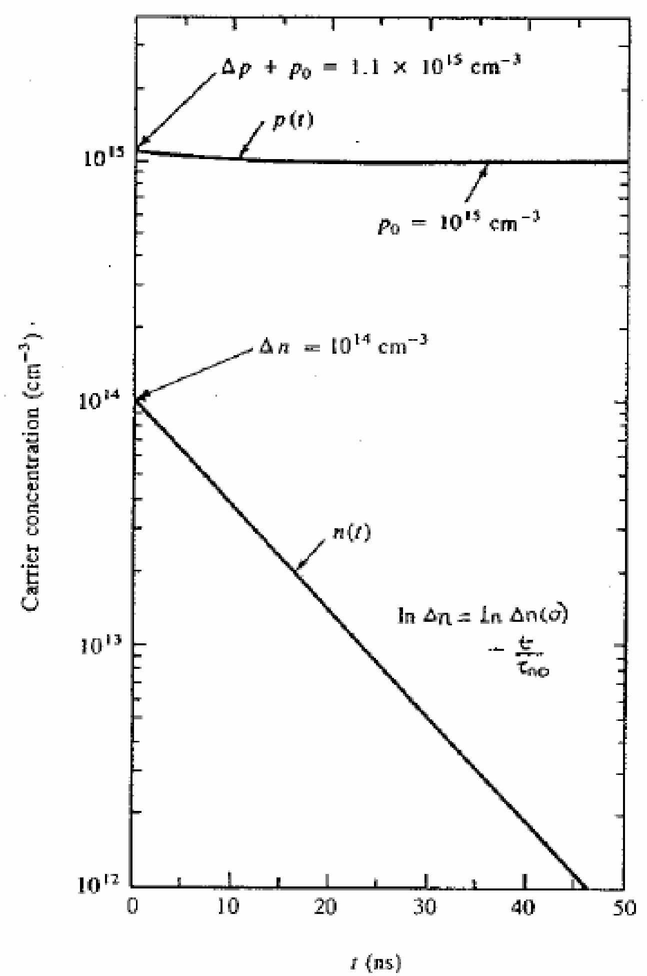


Fig. 7.4

The thermal equilibrium minority concentration is

$$n_0 = \frac{n_i^{\ 2}}{p_0} = \frac{\left(1.5\times10^{10}\right)^2}{10^{15}} = 2.25\times10^5\,/\mathrm{cm}^3$$

From the values of Δn and Δp calculated at various t (10 ns, 20 ns and 30 ns), we see that for low level injection there is large percentage change in the minority carrier concentration (see n_0 value), but only minute percentage change in majority carrier (see p_0 value)

7.2 Continuity Equations

In section 7.1, we have just discussed the change of carrier concentrations due to the optical excitation. We will now analyse the change of carrier concentrations in certain region of a sample as a function of space and time due to the external excitation such as optical excitation and the current flowing into and out of the region.

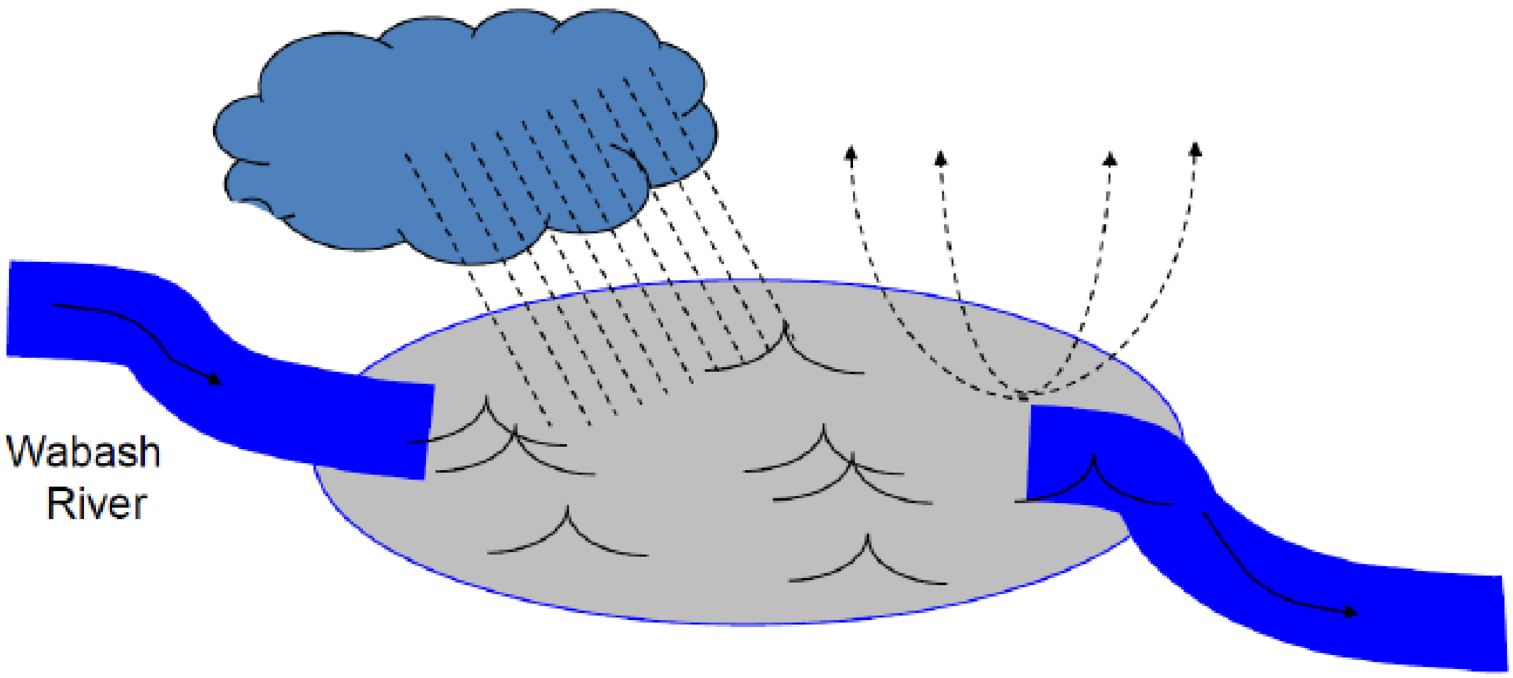
The change of the carrier concentrations in certain region of a sample, say in a volume element *dx dy dz* shown in Fig. 7.5, is governed by

- The change of the excess carrier concentrations due to the generation (thermal generation and external excitation) and recombination, which we just discussed in section 7.1.

- The current flowing in and out of the volume element which may change the carrier concentration

- The total change of the carrier concentrations is then simply the sum of the carrier concentrations change due to the generation, recombination and the current flowing.

Continuity equation



Rate of increase of water level in lake = (in flow - outflow) + rain - evaporation

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \bullet J_n \quad + g_N \quad - r_N$$

We will now evaluate the change of the carrier concentrations due to the current flowing.

Consider a hole current flowing in the positive x-direction into and out the a volume element $dx\ dy\ dz$ as shown in Fig. 7.5

- At x, a hole current density $J_p(x)$ flowing into the volume element.

  The number of holes flowing into the volume element per second is just

$$\Gamma_p(x) = \frac{J_p(x)}{q} dy\ dz \qquad \textbf{(7.37.a)}$$

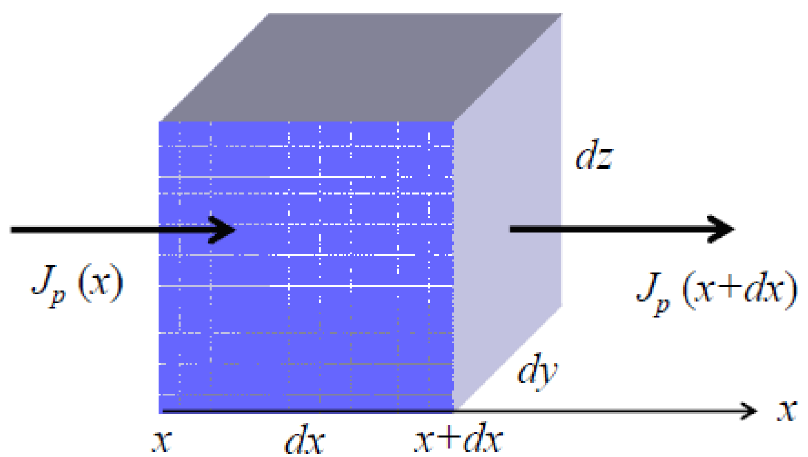


Fig. 7.5

At $(x + dx)$, a hole current density $J_p(x+dx)$ flowing out of the volume element.

The number of holes flowing out of the volume element per second is

$$\Gamma_p(x+dx) = \frac{J_p(x+dx)}{q} dy\, dz \qquad \textbf{(7.37.b)}$$

Using a Taylor series expansion and keeping the first two terms only, we can write

$$J_p(x+dx) = J_p(x) + \frac{\partial J_p(x)}{\partial x} dx \qquad \textbf{(7.37.c)}$$

Hence, eqn. (7.37.b) becomes

$$\Gamma_p(x+dx) = \frac{1}{q}\left(J_p(x) + \frac{\partial J_p(x)}{\partial x} dx \right) dy\ dz \qquad \textbf{(7.37.d)}$$

- From eqns. (7.37.a) and (7.37.d), the increase of the hole concentration in the volume element (*dx dy dz*) per second is

$$\frac{\partial p}{\partial t} dx\ dy\ dz = \Gamma(x) - \Gamma(x+dx)$$

$$\frac{\partial p}{\partial t} dx\ dy\ dz = \frac{1}{q}\left[\left\{ J_p(x) \right\} - \left\{ J_p(x) + \frac{\partial J_p(x)}{\partial x} dx \right\} \right] dy\ dz$$

$$\frac{\partial p}{\partial t} dx\ dy\ dz = -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} dx\ dy\ dz \qquad \textbf{(7.38)}$$

Hence, the increase of the hole concentration per second due to the hole current density is

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p(x)}{\partial x} \qquad \textbf{(7.39)}$$

- Combining the generation, recombination of holes and the flow of holes, the net change of the hole concentration is

$$\boxed{\frac{\partial p(x,t)}{\partial t} = G_{th} + G_L - R - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}} \qquad \textbf{(7.40)}$$

here the first and second terms represent the thermal generation and the generation by the external excitation of holes respectively, the third term represents the recombination and the last term represents the flow of the holes

Similarly, for the case of electrons, we have

$$\boxed{\frac{\partial n(x,t)}{\partial t} = G_{th} + G_L - R + \frac{1}{q}\frac{\partial J_n(x)}{\partial x}} \qquad \textbf{(7.41)}$$

Note the positive (+ ve) sign in front of the $\partial J_n(x)/\partial x$

- Eqns. (7.40) and (7.41) are called the continuity equations for holes and electrons respectively

Consider now a simple case of steady-state low injection from one side of an semi-infinite n-type semiconductor. This is shown schematically in Fig. 7.6.

- Excess carriers are injected at one side of the sample by an external excitation (in this example by an optical excitation). No external excitation at the bulk of the sample. At $x = 0$, the excess minority carrier hole concentration is $\Delta p(0)$.

  $\Delta p(0) \ll n_0$ since this is a low injection case.

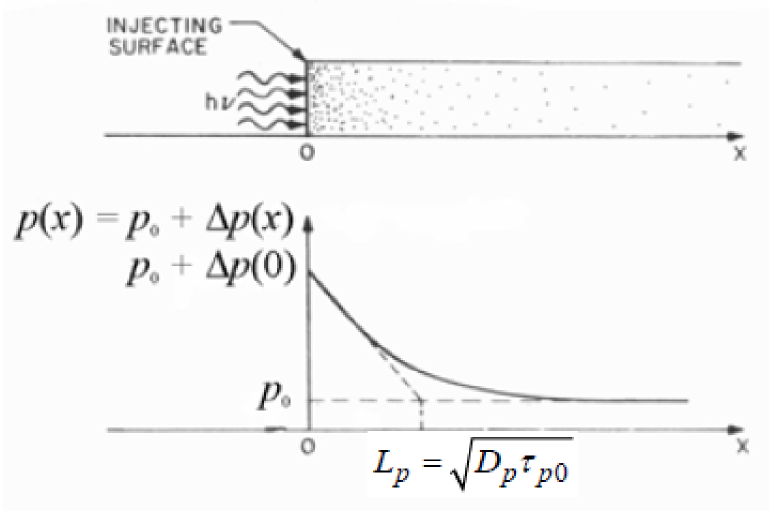


Fig. 7.6

- At the steady state (the carrier concentrations are constant with time) there is a concentration gradient which give a diffusion current. The current is allowed to flow in the sample. At $x = \infty$, the excess minority carrier holes have all recombined and therefore the excess minority carrier hole concentration is zero, or $\Delta p(x = \infty) = 0$

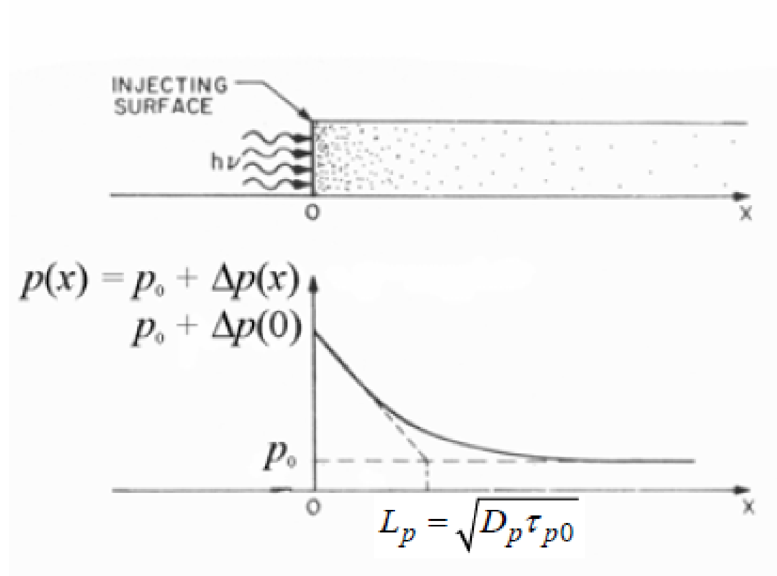


Fig. 7.6

- There is no electric field applied in the sample

- The minority carrier hole concentration, and hence the diffusion current, can be obtained by applying the hole continuity equation [eqn. (7.40)] to this problem

$$\frac{\partial p(x,t)}{\partial t} = G_{th} + G_L - R - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}$$

- Since there is no external excitation in the bulk of the sample $G_L = 0$. The generation of carriers is by thermal generation only.

- For n-type semiconductor where $n_0 >> p_0$ and also for low injection ($\Delta p << n_0$), we have from section 7.1 [see eqn. (7.35)]

$$\frac{dp}{dt} = \frac{d\Delta p}{dt} = G_{th} - R = \alpha_r(n_i^2 - np) = -\frac{\Delta p}{\tau_{p0}}$$

- Since there is no electric field in the sample, the only current is the diffusion current, therefore

$$J_p(x) = -qD_p \frac{\partial p}{\partial x}$$

- Hence, the hole continuity equation for this problem becomes

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\Delta p}{\tau_{p0}} - \frac{1}{q}\frac{\partial}{\partial x}\left(-qD_p \frac{\partial p}{\partial x}\right)$$

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\Delta p}{\tau_{p0}} + D_p \frac{\partial^2 p}{\partial x^2} \qquad \textbf{(7.42)}$$

- In the steady state condition, the carrier concentrations are constant with time, so

$$\frac{\partial p(x,t)}{\partial t} = 0$$

- Therefore eqn.(7.42) becomes $$-\frac{\Delta p}{\tau_{p0}} + D_p \frac{\partial^2 p}{\partial x^2} = 0 \qquad \textbf{(7.43)}$$

Since $p = p_0 + \Delta p$, we can write $$D_p \frac{\partial^2 (p_0 + \Delta p)}{\partial x^2} - \frac{\Delta p}{\tau_{p0}} = 0$$

$$\frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{D_p \tau_{p0}} = 0 \qquad \textbf{(7.44)}$$

- Equation (7.44) is a second order differential equation. The general solution to eqn. (7.44) is

$$\Delta p(x) = A \exp\left(\frac{x}{\sqrt{D_p \tau_{p0}}}\right) + B \exp\left(-\frac{x}{\sqrt{D_p \tau_{p0}}}\right)$$

In this problem (see Fig. 7.6, pages 7-40 and 7-41) we have the following boundary conditions:

- at $x = 0$, $\Delta p(x = 0) = \Delta p(0)$ is a constant, which is the excess minority carrier hole concentration at $x = 0$
- at $x = \infty$, the excess minority carriers have recombined and disappeared, therefore $\Delta p(x = \infty) = 0$

The constants A and B can then be obtained from these two boundary conditions.

- The solution of equation is

$$\boxed{\begin{aligned} &\Delta p(x) = \Delta p(0)\exp\left(-\frac{x}{\sqrt{D_p\,\tau_{p0}}}\right) && \textbf{(7.45.a)} \\ &\Delta p(x) = \Delta p(0)\exp\left(-\frac{x}{L_p}\right) && \textbf{(7.45.b)} \\ &L_p = \sqrt{D_p\,\tau_{p0}} && \textbf{(7.46)} \end{aligned}}$$

with

where $\Delta p(0)$ is the excess minority carrier hole concentration at $x = 0$ and L_p is called the diffusion length for holes

- We see that the excess minority carrier hole concentration falls exponentially inside the semiconductor because of diffusion and recombination

Similarly, for p-type semiconductors (with the same conditions as in the previous case, of course) the excess minority carrier electron concentration is

$$\Delta n(x) = \Delta n(0)\exp\left(-\frac{x}{\sqrt{D_n\,\tau_{n0}}}\right) \quad \textbf{(7.47.a)}$$

$$\Delta n(x) = \Delta n(0)\exp\left(-\frac{x}{L_n}\right) \quad \textbf{(7.47.b)}$$

with

$$L_n = \sqrt{D_n\,\tau_{n0}} \quad \textbf{(7.48)}$$

where $\Delta n(0)$ is the excess minority carrier electron concentration at $x = 0$ and L_n is the diffusion length for electrons

- It can be shown that the minority carrier diffusion length is the average distance a minority carrier diffuses before recombining. It also represents the distance at which the steady state excess concentration decays to 1/e of its value

7.3 Excess Carrier Concentrations (Indirect Transitions)

- At this point we should note that our analysis on the excess minority carriers in the previous sections is based on the direct band-to-band transitions. We have considered an ideal semiconductor in which there is no defects energy states in the bandgap.

- In chapter 3.4.3, we have learnt that for indirect bandgap semiconductors, in the presence of defect states, it is more likely that the electrons in the conduction band recombine with holes in the valence band through the defect states in the bandgap. The defect states, also called trap, act as recombination centers capturing both electrons and holes. This type of recombination is called indirect recombination.

  It turns out that the expressions of the excess minority carrier concentration for the indirect transitions are identical to those for the direct transitions. The only difference is in the expression for the minority carrier lifetime.

- For direct band-to-band transitions, the minority carrier lifetime is given by

$$\tau_{n0} = \frac{1}{\alpha_r \, p_0} \qquad (7.19) \qquad\qquad \tau_{p0} = \frac{1}{\alpha_r \, n_0} \qquad \mathbf{(7.23)}$$

- For indirect transitions (see Appendix C), the minority carrier lifetime is given by

$$\tau_{n0} = \frac{1}{C_n N_t} \qquad (C.11) \qquad\qquad \tau_{p0} = \frac{1}{C_p N_t} \qquad (C.20)$$

  where C_n and C_p are constants proportional to electron-capture and hole-capture cross sections respectively, and N_t is trap concentration.

- The analysis for the indirect transition processes is discussed in Appendix C. Here, we summarize the excess minority carrier concentration for the direct band-to-band transitions and the indirect transitions.

Summary of the excess minority carrier concentration of p-type semiconductors where $p_0 >> n_0$ and under low injection $(p_0 >> \Delta n)$. For the indirect transitions, the trap energy level is assumed to be near the midgap, thus $E_t \approx E_i$

	Direct band-to-band transitions	Indirect transitions
General equation for minority carrier	$\frac{dn}{dt} = \frac{d\Delta n}{dt} = G_{th} + G_L - R = G_L - \frac{\Delta n}{\tau_{n0}}$ **(7.9) and (7.26) to (7.29), or (C.13)**	$\frac{dn}{dt} = \frac{d\Delta n}{dt} = G_L - R' = G_L - \frac{\Delta n}{\tau_{n0}}$ **(C.12)**
The excess minority carrier lifetime	$\tau_{n0} = \frac{1}{\alpha_r p_0}$ **(7.19) or (7.30)**	$\tau_{n0} = \frac{1}{C_n N_t}$ **(C.11)**
Steady state under illumination with generation rate G_L $(dn/dt = 0)$	$G_L = \alpha_r\ p_0\ \Delta n = \frac{\Delta n}{\tau_{n0}}$ $\Delta n = G_L \tau_{n0}$ **(7.18)**	$G_L = \frac{\Delta n}{\tau_{n0}}$ $\Delta n = G_L\ \tau_{n0}$ **(C.14)**
After illumination is switched off, $G_L = 0$ and $dn/dt \neq 0$	$\Delta n(t) = \Delta n(0) \exp\left(-\frac{t}{\tau_{n0}}\right)$ **(7.33)**	$\Delta n(t) = \Delta n(0) \exp\left(-\frac{t}{\tau_{n0}}\right)$ **(C.15)**

Summary of the excess minority carrier concentration of p-type semiconductors where $p_0 >> n_0$ and under low injection $(p_0 >> \Delta n)$. For the indirect transitions, the trap energy level is assumed to be near the midgap, thus $E_t \approx E_i$

	Direct band-to-band transitions	Indirect transitions
General continuity equation for electrons	$\frac{\partial n(x,t)}{\partial t} = G_{th} + G_L - R + \frac{1}{q}\frac{\partial J_n(x)}{\partial x}$ $\frac{\partial n(x,t)}{\partial t} = G_L - \frac{\Delta n}{\tau_{n0}} + \frac{1}{q}\frac{\partial J_n(x)}{\partial x}$ **(7.41)**	$\frac{\partial n(x,t)}{\partial t} = G_L - R' + \frac{1}{q}\frac{\partial J_n(x)}{\partial x}$ $\frac{\partial n(x,t)}{\partial t} = G_L - \frac{\Delta n}{\tau_{n0}} + \frac{1}{q}\frac{\partial J_n(x)}{\partial x}$ **(C.16)**
Steady state injection of excess minority carriers at one end of semi-infinite p-type semiconductor (similar to Fig.7.6). No electric field. Boundary conditions: at $x = 0$, $\Delta n\ (x = 0) = \Delta n\ (0)$ at $x = \infty$, $\Delta n\ (x = \infty) = 0$.	$\Delta n(x) = \Delta n(0)\exp\left(-\frac{x}{L_n}\right)$ **(7.47.b)** $L_n = \sqrt{D_n \tau_{n0}}$ **(7.48)**	$\Delta n(x) = \Delta n(0)\exp\left(-\frac{x}{L_n}\right)$ **(C.17)** $L_n = \sqrt{D_n \tau_{n0}}$ **(C.18)**

Summary of the excess minority carrier concentration of n-type semiconductors where $n_0 >> p_0$ and under low injection $(n_0 >> \Delta p)$. For the indirect transitions, the trap energy level is assumed to be near the midgap, thus $E_t \approx E_i$

	Direct band-to-band transitions	Indirect transitions
General equation for minority carrier	$\frac{dp}{dt} = \frac{d\Delta p}{dt} = G_{th} + G_L - R = G_L - \frac{\Delta p}{\tau_{p0}}$ **(7.9) and (7.35)**	$\frac{dp}{dt} = \frac{d\Delta p}{dt} = G_L - R' = G_L - \frac{\Delta p}{\tau_{p0}}$ **(C.19)**
The excess minority carrier lifetime	$\tau_{p0} = \frac{1}{\alpha_r n_0}$ **(7.23)**	$\tau_{p0} = \frac{1}{C_p N_t}$ **(C.20)**
Steady state under illumination with generation rate G_L $(\mathrm{d}n/\mathrm{d}t = 0)$	$G_L = \alpha_r\, n_0\, \Delta p = \frac{\Delta p}{\tau_{p0}}$ $\Delta p = G_L\, \tau_{p0}$ **(7.22)**	$G_L = \frac{\Delta p}{\tau_{p0}}$ $\Delta p = G_L\, \tau_{p0}$ **(C.21)**
After illumination is switched off, $G_L = 0$ and $\mathrm{d}n/\mathrm{d}t \neq 0$	$\Delta p(t) = \Delta p(0)\exp\left(-\frac{t}{\tau_{p0}}\right)$ **(7.36)**	$\Delta p(t) = \Delta p(0)\exp\left(-\frac{t}{\tau_{p0}}\right)$ **(C.22)**

Summary of the excess minority carrier concentration of n-type semiconductors where $n_0 >> p_0$ and under low injection $(n_0 >> \Delta p)$. For the indirect transitions, the trap energy level is assumed to be near the midgap, thus $E_t \approx E_i$

	Direct band-to-band transitions	Indirect transitions
General continuity equation for electrons	$\frac{\partial p(x,t)}{\partial t} = G_{th} + G_L - R - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}$ $\frac{\partial p(x,t)}{\partial t} = G_L - \frac{\Delta p}{\tau_{p0}} - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}$ **(7.40)**	$\frac{\partial p(x,t)}{\partial t} = G_L - R' - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}$ $\frac{\partial p(x,t)}{\partial t} = G_L - \frac{\Delta p}{\tau_{p0}} - \frac{1}{q}\frac{\partial J_p(x)}{\partial x}$ **(C.23)**
Steady state injection of excess minority carriers at one end of semi-infinite p-type semiconductor (see Fig.7.6). No electric field. Boundary conditions: at $x = 0$, $\Delta p\ (x = 0) = \Delta p\ (0)$ at $x = \infty$, $\Delta p\ (x = \infty) = 0$.	$\Delta p(x) = \Delta p(0)\exp\left(-\frac{x}{L_p}\right)$ **(7.45.b)** $L_p = \sqrt{D_p\ \tau_{p0}}$ **(7.46)**	$\Delta p(x) = \Delta p(0)\exp\left(-\frac{x}{L_p}\right)$ **(C.24)** $L_p = \sqrt{D_p\ \tau_{p0}}$ **(C.25)**

Example 7.2

A semi-infinite Si sample is doped uniformly with phosphorus atoms to a concentration $N_d = 10^{17}/\text{cm}^3$. Excess minority carriers are generated at one end of the sample ($x = 0$) to a concentration of $10^{15}/\text{cm}^3$ at the steady state. The minority carrier lifetime is 1 μs. The electron and hole diffusion coefficient is 25 cm^2/s and 10 cm^2/s. The intrinsic carrier concentration at room temperature is $10^{10}/\text{cm}^3$. Determine the steady state excess minority carrier concentration at x = 10 μm.

Solution: $N_d >> n_i$ and assume the donor atoms are fully ionized at room temperature. Therefore the thermal equilibrium electron concentration $n_0 = N_d = 10^{17}/\text{cm}^3$.

The excess minority carrier concentration at $x = 0$ is $\Delta p\,(0) = 10^{15}/\text{cm}^3$. We see that $\Delta p\,(0) << n_0$, therefore it is low injection.

The excess minority carrier is holes. The determine the excess minority carrier concentration, we simply use eqn. (7.45.b) or eqn. (C.24), they are the same.

Recall eqn. (7.45.b) or eqn. (C.24)

$$\Delta p(x) = \Delta p(0)\exp\left(-\frac{x}{L_p}\right)$$

We first calculate L_p using eqn. (7.46) or eqn. (C.25)

$$L_p = \sqrt{D_p\ \tau_{p0}} = \sqrt{10\times10^{-6}} = 3.16\times10^{-3}\ \text{cm}$$

$$\Delta p(x) = 10^{15}\exp\left(-\frac{x}{3.16\times10^{-3}}\right)\ /\text{cm}^3$$

At x = 10 μm, $$\Delta p(10\ \mu\text{m}) = 10^{15}\exp\left(-\frac{10\times10^{-4}}{3.16\times10^{-3}}\right)\ /\text{cm}^3$$

$$\Delta p(10\ \mu\text{m}) = 7.3\times10^{14}/\text{cm}^3$$

7.4 Quasi Fermi Level

Recall the thermal equilibrium electron and hole concentrations as functions of the Fermi energy levels described by eqns. (5.37) and (5.38)

$$n_0 = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right] \qquad \textbf{(5.37)}$$

$$p_0 = n_i \exp\left[\frac{E_i - E_F}{k_B T}\right] \qquad \textbf{(5.38)}$$

- For an n-type semiconductor $E_F > E_i$, and we get $n_0 > n_i$ and $p_0 < n_i$. E_F increases with increasing n_0
- For a *p*-type semiconductor $E_F < E_i$, and we get $n_0 < n_i$ and $p_0 > n_i$. E_F decreases with increasing p_0. (see Fig. 7.7)

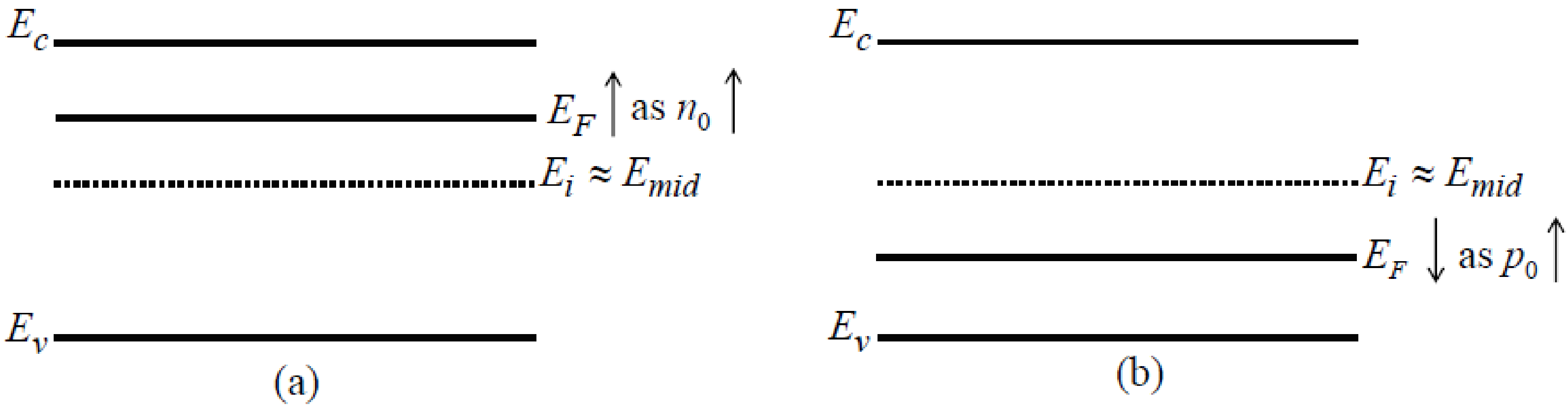


Fig. 7.7 Thermal equilibrium energy band diagram for (a) n-type semiconductor and (b) p-type semiconductor

With excess carriers created in the semiconductor, we are no longer in thermal equilibrium

- The electron and hole concentrations are no longer the thermal equilibrium electron and hole concentrations. The Fermi energy level for the thermal equilibrium case in eqns. (5.37) and (5.38) can no longer be used.

- However, we can define a quasi-Fermi level for electrons and a quasi Fermi level for holes for the non-equilibrium condition.

- Again, as in the case of thermal equilibrium, these quasi-Fermi levels are determined by the non-thermal equilibrium carrier concentrations, n and p. The equations take similar form as eqns. (5.37) and (5.38) [the equations for the thermal equilibrium carrier concentrations] and are written as follow

$$\boxed{\begin{aligned} n = n_0 + \Delta n &= n_i \exp\left[\frac{E_{Fnq} - E_i}{k_B T}\right] \quad \textbf{(7.49.a)} \\ p = p_0 + \Delta p &= n_i \exp\left[\frac{E_i - E_{Fpq}}{k_B T}\right] \quad \textbf{(7.49.b)} \end{aligned}}$$

where E_{Fnq} and E_{Fpq} are the quasi-Fermi energy levels for electrons and holes respectively.

- The behavior of the quasi-Fermi levels with the excess carrier concentration is best described using example 7.3 below.

Example 7.3

Consider an n-type semiconductor at 300 K with carrier concentrations of $n_0 = 10^{15}$ /cm^3, $n_i = 10^{10}$ /cm^3. In non-equilibrium, the excess carrier concentrations become $\Delta n = \Delta p = 10^{13}$ /cm^3. Determine the initial Fermi level under thermal equilibrium and the quasi-Fermi levels in non-equilibrium, and indicate these levels in a band diagram.

From eqn. (5.31), the minority carrier concentration is

$$p_0 = \frac{n_i^{\,2}}{n_0} = \frac{\left(10^{10}\right)^2}{10^{15}} = 10^5 \text{ /cm}^3$$

The Fermi energy level for thermal equilibrium can be determined from either eqns. (5.37) or (5.38).

From eqn. (5.37) $$E_F - E_i = k_B T \ln\left(\frac{n_0}{n_i}\right) = 0.2982 \text{ eV}$$

From eqn. (5.38) $$E_i - E_F = k_B T \ln\left(\frac{p_0}{n_i}\right) = -0.2982 \text{ eV}$$

the position of E_F is the same

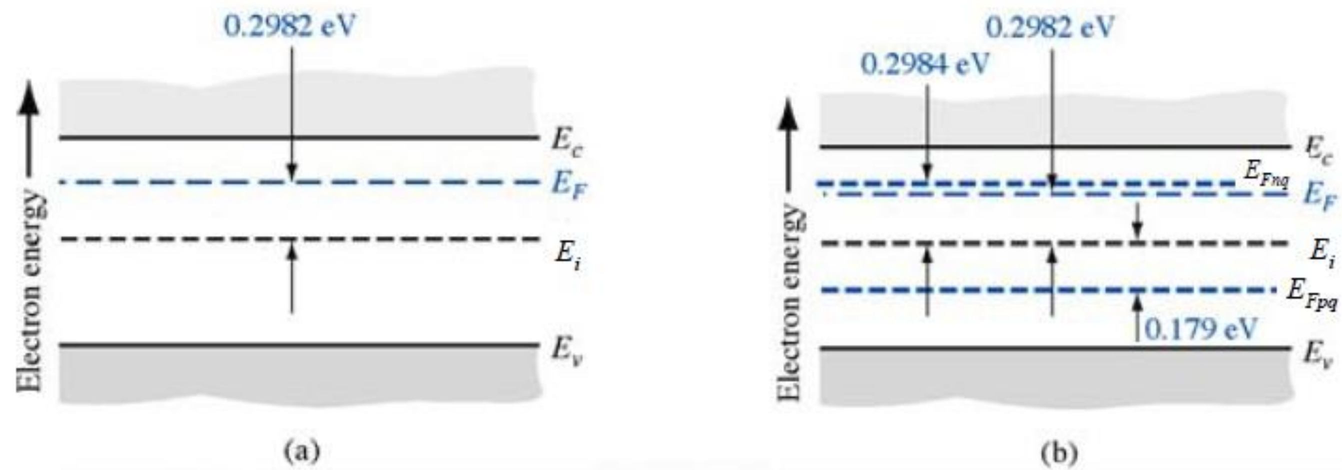
The quasi-Fermi energy level for electrons in non-equilibrium is determined from eqn. (7.49.a)

$$E_{Fnq} - E_i = k_B T \ln\left(\frac{n_0 + \Delta n}{n_i}\right) = 0.2984 \text{ eV}$$

The quasi- Fermi energy level for holes in non-equilibrium is determined from eqn. (7.49.b)

$$E_i - E_{Fpq} = k_B T \ln\left(\frac{p_0 + \Delta p}{n_i}\right) = 0.179 \text{ eV}$$

The energy band diagram is shown in Fig. 7.8 below



(a) Thermal-equilibrium energy-band diagram for $N_d = 10^{15}$ cm^{-3} and $n_i = 10^{10}$ cm^{-3}
(b) Quasi-Fermi levels for electrons and holes if 10^{13} cm^{-3} excess carriers are present

Fig. 7.8

From the above example we note that

- The quasi-Fermi level for electrons is not much different from the thermal equilibrium Fermi level. This is because, for low injection ($\Delta n << n_0$), the majority carrier electron concentration does change significantly.

- On the other hand, the hole concentration has increased significantly since $\Delta p >> p_0$. Therefore, the quasi-Fermi level for holes deviates significantly from the Fermi level.

In addition, any drift, diffusion or combination of the two in a semiconductor results in currents proportional to the gradient of the quasi-Fermi levels. When there is no current current flowing, the Fermi level is constant throughout the sample